

KARTHIK JONNALAGADDA

9392991430 | jonnalagadda.karthik22@st.niituniversity.in | [linkedin.com/in/jonnalagadda-karthik01](https://www.linkedin.com/in/jonnalagadda-karthik01) | github.com/karthikjonnalagadda | karthikjonnalagadda.github.io

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building and deploying machine learning systems in Python. Built Artificial Intelligence applications covering Retrieval-Augmented Generation (RAG), Computer Vision, NLP, and OCR, served through FastAPI REST APIs backed by MongoDB and vector databases (ChromaDB, FAISS). Works across the full ML lifecycle - data preprocessing, feature engineering, model training, evaluation, and deployment - and designs systems that keep running when a model or dependency fails. Strong problem-solving and testing habits from enterprise QA work: automated test suites, SQL data validation, and root-cause debugging in Agile teams.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, TypeScript, Java, R

Machine Learning: Scikit-learn, Classification, Regression, Clustering (K-Means), Dimensionality Reduction (SVD, NMF), Feature Engineering, Data Preprocessing, Model Evaluation (ROC-AUC, Precision/Recall/F1, ARI)

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Statistics, Data Cleaning, Data Visualization

Deep Learning & AI: PyTorch, Transformers, Neural Networks, Convolutional Neural Networks (DenseNet121), Computer Vision, Natural Language Processing (NLP), OCR, Explainable AI (Grad-CAM, SHAP, LIME)

LLM & Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Sentence Transformers, Embeddings, Vector Databases (ChromaDB, FAISS), LangChain, Hybrid Retrieval

Backend & Databases: FastAPI, Flask, REST API Design, Node.js, MongoDB, MySQL, Streamlit, React

Tools & DevOps: Git, GitHub, Docker, Jenkins (CI/CD), Linux, Postman, pytest, AWS (EC2), Agile

QA & Analytics: Selenium, Playwright, API Testing, Regression Testing, SQL Validation, Root Cause Analysis, Power BI

PROJECTS

Medical Multimodal AI Assistant - RAG, Computer Vision & OCR | NIIT University

Python, FastAPI, PyTorch, Sentence Transformers, ChromaDB, FAISS, LangChain, EasyOCR, Tesseract, MongoDB, React, Streamlit

- Architected a multimodal clinical AI system exposing **10 REST endpoints** over FastAPI, serving OCR, X-ray analysis, RAG diagnosis, and chat to a React dashboard and a Streamlit UI; deferred model loading off the request path to protect API responsiveness.
- Engineered a provider-agnostic LLM layer over **4 backends** (Groq, OpenAI, xAI, local BioGPT) and a **3-backend** vision registry that degrades gracefully when model weights are unavailable - keeping the service running instead of failing when a dependency is missing.
- Trained a **7.49M-parameter DenseNet121** for **14-class** NIH Chest X-ray multi-label classification with per-class ROC-AUC evaluation, and grounded LLM output in retrieved clinical evidence via hybrid dense + lexical retrieval over ChromaDB/FAISS.
- Diagnosed a schema-contract defect between prompt format and enforced JSON response mode that made parsing unreliable; imposed a strict clinical schema with a normalization layer and locked it behind regression tests. Eliminated duplicate vector writes by making ingestion idempotent with content-hash chunk IDs.
- Hardened the codebase to **50 automated tests passing (100%) at 60% coverage** with pre-commit, flake8, and pylint gates, and split a minimal runtime dependency set from the development set to cut install surface.

SafeInbox - Real-Time Email Spam Detection | NIIT University

Python, Scikit-learn, TF-IDF, Node.js, Express, React, MongoDB, Docker, Jenkins

- Trained a Naive Bayes spam classifier on **26,972 labeled Enron emails** across a **1.03M-feature TF-IDF** space, persisting model and vectorizer artifacts for real-time inference.
- Bridged the scikit-learn model into a Node/Express runtime through a Python inference process, avoiding a separate model-serving service, and ingested live mail through both the Gmail API (OAuth2) and IMAP.
- Secured the platform with JWT authentication, bcrypt hashing, and protected routes; containerized with Docker and automated builds through a Jenkins CI/CD pipeline.

Genomic Data Dimensionality Reduction & Clustering | NIIT University

Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

- Built a reproducible unsupervised learning pipeline on high-dimensional genomic data, benchmarking two dimensionality-reduction strategies - SVD and NMF - ahead of K-Means clustering, and selected SVD on measured cluster agreement: **Adjusted Rand Index 0.66 (SVD) vs 0.56 (NMF)**.
- Validated cluster quality independently of the clustering objective by training a Random Forest classifier on the reduced feature space, confirming the groups were separable rather than artifacts of the projection.
- Drove exploratory data analysis (EDA) to interpret clusters biologically - highly variable gene identification, correlation heatmaps, and top-gene visualization in Pandas, NumPy, Matplotlib, and Seaborn.

EXPERIENCE

QA Automation Engineer Intern - NIIT Ltd (neo.niit.com) | Jan 2026 - Present

- Automated regression, functional, and cross-browser suites in Selenium WebDriver across high-traffic revenue workflows, cutting manual test effort and widening release coverage.
- Validated backend SQL/MySQL integrity between application actions and persisted records, intercepting data-quality defects before production release, and verified REST API responses with Postman alongside Power BI / Zoho Analytics reporting audits.
- Drove structured root-cause analysis with developers across Agile sprints, converting defect investigations into documented test cases and release sign-off.

EDUCATION

B.Tech, Computer Science Engineering (Data Science) - NIIT University, Neemrana | 2022 - 2026

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Data Science Professional - Oracle